

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 10

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Topic

Optical and Instrument Calibration

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Optical and Instrument Calibration. State all assumptions and conventions.

$$\gamma(\nu) = \gamma + \gamma_c(\nu - \nu_0)/\nu_0$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$c_{read} = \lambda c_m$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$C_{opt} = \begin{pmatrix} \sigma_\gamma^2 & \cdots \\ \vdots & \sigma_\lambda^2 \end{pmatrix}$$

Show enough intermediate work that another student can reproduce the calculation.